

Agentic AI Roadmap

1. Python Foundation

Learn:

- OOP
- APIs
- JSON
- Async programming
- Virtual environments
- FastAPI
- Pydantic

Projects:

- CLI chatbot
 - REST API
 - File assistant
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2. LLM Fundamentals

Learn:

- Tokens
- Context windows
- Embeddings
- Transformers
- Prompt engineering
- Hallucination
- Function calling

Use:

- Ollama
- OpenRouter
- OpenAI APIs

Projects:

- Chatbot
 - PDF Q&A bot
 - Code assistant
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3. LangChain Ecosystem

Learn:

- Prompt templates
- Chains
- Memory
- Tools
- Agents
- Output parsers

Frameworks:

- LangChain
- LangGraph

Projects:

- Weather agent
 - Calculator agent
 - Web search agent
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4. RAG Systems

Learn:

- Vector databases
- Chunking
- Embeddings
- Retrieval pipelines
- Re-ranking

Tools:

- ChromaDB
- FAISS

Projects:

- ERP knowledge bot
 - Document search assistant
 - Notes assistant
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5. Agentic AI Core

Learn:

- Tool calling
- Reflection
- Planning
- Task decomposition
- Multi-agent workflows
- Memory systems

Frameworks:

- LangGraph
- CrewAI
- AutoGen

Projects:

- Research agent
 - Coding agent
 - Autonomous debugging assistant
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6. Infrastructure

Learn:

- Docker

- PostgreSQL
- Redis
- FastAPI deployment
- Streaming APIs
- Logging and tracing

Projects:

- Deploy AI services
 - Build production APIs
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7. Advanced Topics

Learn:

- MCP
- Voice agents
- Fine tuning
- Local models
- Multimodal AI
- Agent orchestration

Projects:

- Browser automation agent
 - Voice assistant
 - Autonomous workflow system
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Recommended Stack

Frontend:

- Next.js

Backend:

- FastAPI

AI:

- LangChain
- LangGraph

Models:

- Ollama
- OpenRouter

Database:

- PostgreSQL

Vector DB:

- ChromaDB

Deployment:

- Docker
- Railway
- Render

Important Advice

- Build projects continuously.
- Do not stay in tutorial hell.
- Learn RAG before autonomous agents.
- Focus on production systems.
- Understand fundamentals before frameworks.